Relative chains are defined as the cokernel of the singular-chain map induced by the inclusion of the relative subset into the ambient subset. Relative homology is the homology of this complex in the selected degree. These constructions do not yet supply the relative maps, connecting morphisms, excision, or the well-group theorem.

Result 1 (chain cokernel kills its defining map). [Topology foundations, CategoryTheory.Limits.cokernel.condition]
Let K be a type. Let *field* be a field structure on K. Let C and D be elements of Ch_K. Let f be a chain map from C to D. Then

$$\pi_f \circ f = 0.$$

Adapted from the cited source.

Result 2 (singular chain homology agrees). [Topology foundations, AlgebraicTopology.singularHomologyFunctor]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Then

$$H_{H.\mathrm{degree}}(C_*(X; H)) = H(X).$$

Result 3 (relative chains are the inclusion cokernel). [Topology foundations, relative singular chain definition]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be two nested subsets of X with inclusion evidence. Then

$$C_*(Q; H) = \mathrm{coker}(C_*(Q.\mathit{relative} \hookrightarrow Q.\mathit{ambient})).$$

Result 4 (relative homology uses the selected degree). [Topology foundations, relative singular homology definition]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be two nested subsets of X with inclusion evidence. Then

$$H(Q) = H_{H.\mathrm{degree}}(C_*(Q; H)).$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.